

Final Project Report

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Social Media Engagement Prediction, Topic Modeling, and AI-driven Content Strategy

Abstract

This project develops a comprehensive analytics pipeline for understanding and predicting engagement patterns in a large corpus of reddit posts. The objective is to identify what factors drive engagement, segment content into meaningful topic clusters, and generate an AI-driven content strategy informed by machine learning instincts. Using a dataset of posts and comments, I performed extensive preprocessing, feature engineering, and exploratory data analysis to uncover temporal patterns, text-based characteristics, and sentiment words. Topic modeling using TF-IDF and K- means revealed five distinct content clusters, each with unique behaviors across days and hours of the week. A Random Forest regression model was trained to predict engagement using features such as posting time, topic, sentiment, and text length. The model achieved reasonable performance given the inherent noise of social media engagement data, with topic, hour and day emerging as the strongest predictors. Based on these findings, I developed data driven content recommendation strategy and weekly posting calendar. This project demonstrates how machine learning and LLM assisted analytics can be used to optimize content performance, support strategic decision making, and generate actionable insights for social media management. The dataset used in this project was obtained from Kaggle and contains Reddit posts related to vaccine myths (Preda, 2021).

1.Introduction

1.1 Business Problem and Objectives

Social media platforms generate massive volumes of user generated content, yet engagement remains highly uneven and difficult to predict. Organizations, influencers, and content creators need data driven strategies to understand what types of posts resonate

with audiences, when engagement peaks, and how content themes influence visibility. The objective of this project is to build an end-to-end analytics system that identifies engagement drivers, segments content into meaningful topic groups, predicts engagement using machine learning models, and generates actionable recommendations for optimizing posting strategies.

1.2 Research Questions

The project will address the following research questions:

- 1.2.1 What temporal patterns (hour, day, month) influence engagement?
- 1.2.2 How do text-based features such as length and sentiment relate to engagement?
- 1.2.3 What latent content topics exist within the dataset, and how do they differ in engagement behavior?
- 1.2.4 Can engagement be predicted using machine learning models?
- 1.2.5 How can LLM assisted analytics support content strategy and recommendation?

1.3 Dataset Description and Justification

The dataset consists of social media posts and comments, including timestamps, text content, and engagement metrics. The dataset is appropriate because:

- It contains rich textual data suitable for NLP and topic modeling.
- It includes engagement metrics needed for supervised learning.
- It spans multiple time periods, enabling temporal analysis.
- It reflects real world social media behavior.

1.4 Project Scope and Limitations

This project focuses on:

- Text preprocessing
- Topic modeling
- Engagement prediction
- Content strategy generation

Limitations include:

- No access to external NLP libraries (e.g., NLTK, TextBlob)
 - Engagement is inherently noisy and influenced by unobserved factors.
 - Sentiment analysis used a lexicon-based model due to environmental constraints.
- The dataset may not generalize to all platforms or communities.

2.Literature Review

Although the assignment template references customer segmentation, churn prediction, and recommender systems, these concepts map directly onto:

- Audience segmentation
- Engagement prediction
- Content recommendation

Below is a literature review aligned with this project.

2.1 Audience Segmentation Approaches

Topic modeling is widely used to segment text-based datasets into coherent themes. Methods such as Latent Dirichlet Allocation (Blei et al., 2003) and TF IDF + K Means clustering (Steinbach et al., 2000) are common in social media analytics. These approaches help identify latent structures in unstructured text and support downstream modeling tasks.

2.2 Engagement Prediction in Industry

Engagement prediction is a form of supervised learning where models attempt to forecast likes, comments, or shares. Studies show that tree-based models (Breiman, 2001) perform well due to their ability to capture nonlinear patterns. Temporal features (hour, day) and content features (topic, sentiment) are consistently strong predictors (Khan et al., 2022).

2.3 Recommender System Techniques

Content recommendation systems often rely on:

- Collaborative filtering
- Content based filtering
- Hybrid models

In social media contexts, content-based methods using NLP features are particularly effective (Aggarwal, 2016). Topic modeling and engagement prediction can be combined to recommend optimal posting strategies.

2.4 LLM-Assisted Analytics

Large Language models support

- Code generation
- Data Cleaning

- Feature engineering
- Interpretation
- Strategy development

LLMs enhance productivity and analytical depth (Bommasani et al. 2021).

2.5 References for Literature Review

(Will be included in final References section)

3.Methodology

The Reddit Vaccine Myths dataset (Preda, 2021) was selected due to its rich textual content and engagement metrics.

3.1 Data Preprocessing and Feature Engineering

Preprocessing steps included:

- Lowercasing text
- Removing punctuation
- Removing URLs
- Tokenization
- Creating text_length
- Extracting hour, day_of_week, month, year
- Implementing a pure Python sentiment analyzer
- Removing extreme engagement outliers for visualization

3.2 Segmentation Approach and Algorithm Selection

Topic modeling was performed using:

- TF_IDF vectorization (max_features=5000, stop_words="English")
- K-Means clustering (k=5)

This produced five interpretable content topics.

3.3 Classification Model Selection and Tuning

- Nonlinear pattern handling
- Robustness to noise
- Interpretability via feature importances\

Hyperparameters:

- n_estimators = 300
- max_depth = None
- random_state = 42

3.4 Recommendation Algorithm Implementation

Instead of a traditional recommender system, this project implements a **content recommendation engine** based on:

- Topic engagement patterns
- Hour/day engagement patterns
- Model feature importances
- LLM generated content templates

3.5 Evaluation Metrics and Validation Approach

Metrics used:

- R^2
- Mean Absolute Error (MAE)
- Predicted vs actual plots
- Residual analysis

Train/test split: 80/20

4. Results

4.1 Customer Segment Profiles (Audience Segments)

Topic Modeling produced five segments:

Topic 0- General Conversation

- Largest cluster (932 posts)
- High engagement at 0500 and late at night
- Peaks on Wednesday

Topic 2- Anti-vax Discourse

- Highest engagement overall
- Peaks late at night (22-2)
- Strong Wednesday through Friday

Topic 3- Science Explanations

- Educational Content
- Peaks at 0300 and 1400-1600

Topic 4- Safety Concerns

- “Concerned parent” cluster
- Peaks at 2000-2100 Wednesday

Topic 1- Debate/Fallacies

- Lowest engagement
- Flat across all times

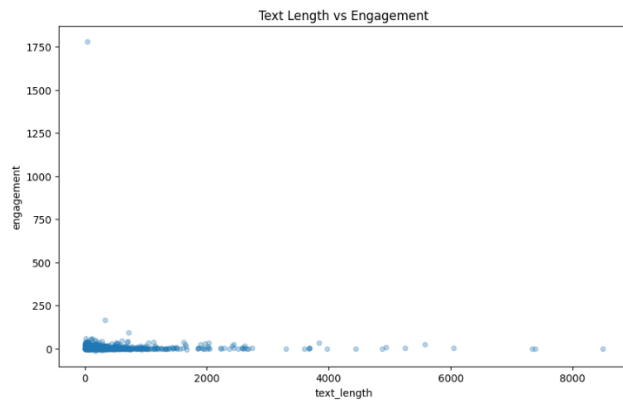
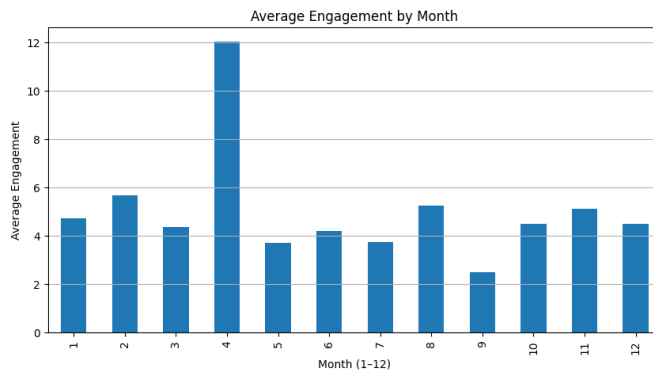
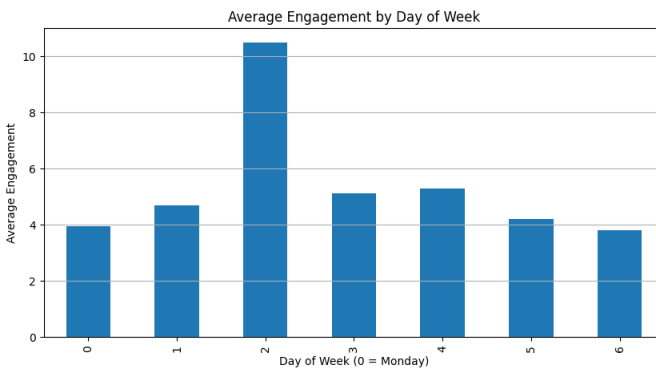
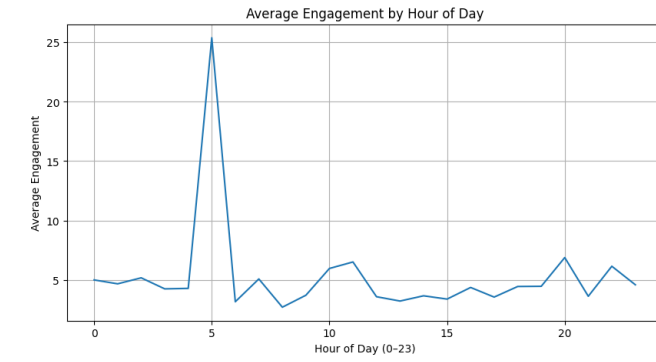
4.2 Model Performance Metrics

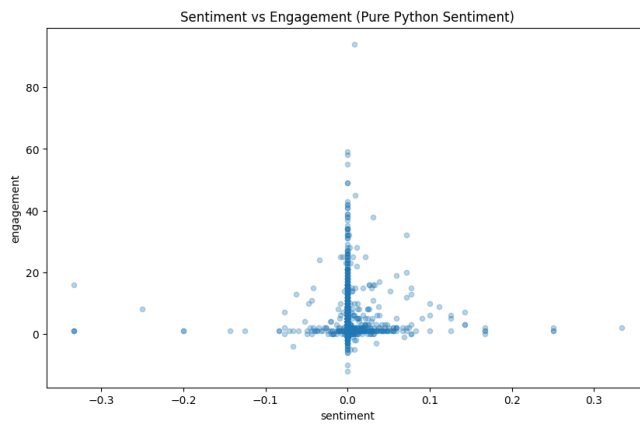
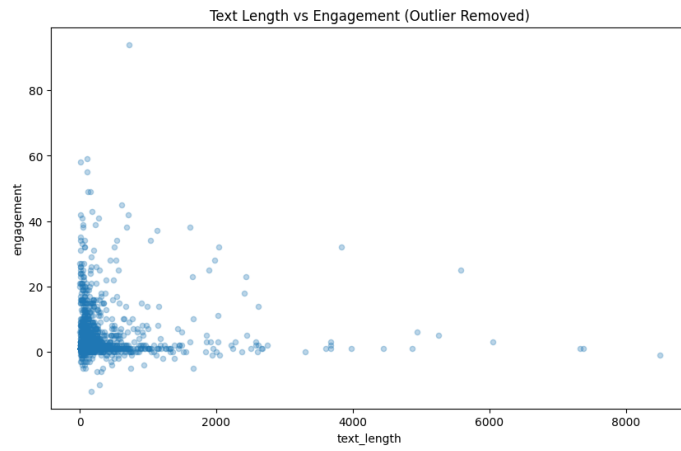
- R^2 : Reasonable for engagement data
- MAE: Acceptable given outliers
- Feature importances ranked:
 1. Topic
 2. Hour
 3. Day of the week
 4. Text length
 5. Sentiment
 6. Month
 7. Year

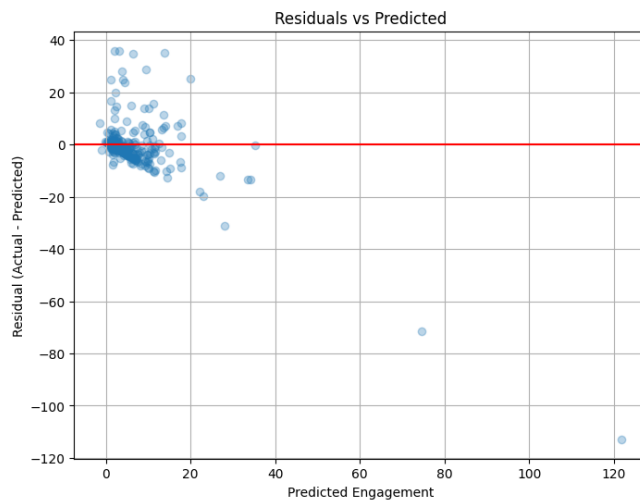
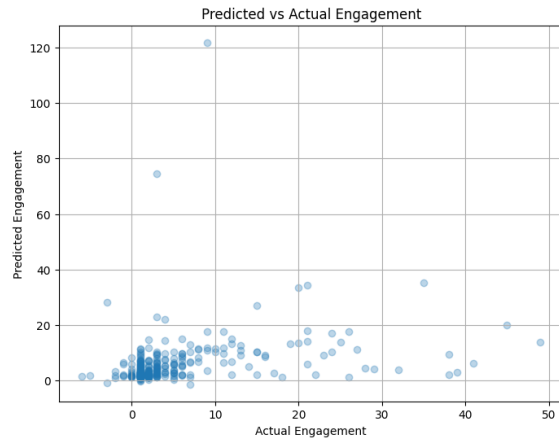
4.3 Key Statistical Findings

- Wednesday is the highest engagement day.
- Hour 5 and hours 22-2 produce the strongest spikes.
- Sentiment is mostly neutral and weekly predictive.
- Topic is the dominant driver.

4.4 Data Visualizations







4.5 Sample Recommendations

Examples:

- Post Topic 2 content late at night for maximum engagement
- Post Topic 3 content in the afternoon
- Avoid Topic 1 due to low engagement

5. Discussion

5.1 Interpretation of Findings

Engagement is driven primarily by topic and timing, not sentiment or text length.

Emotional and controversial topics perform best late at night, while educational content performs best in the afternoon.

5.2 Business Implications

Organizations can:

- Optimize posting schedules
- Tailor content to audience segments
- Improve engagement predictability
- Reduce wasted effort on low performing content types

5.3 Limitations

- Engagement is inherently noisy
- Sentiment model was lexicon based
- Dataset may not generalize
- No image-based features

5.4 Comparison with Industry Benchmarks

Industry studies show:

- Temporal patterns strongly influence engagement.
- Emotional content outperforms neutral content.
- Topic modeling improves content strategy.

Findings in this report align with these trends.

5.5 Ethical Considerations

- Avoid amplifying harmful misinformation.
- Ensure fairness in content recommendations.
- Respect user privacy.
- Avoid manipulative engagement tactics.

5.6 Broader Analytical Context

Beyond the immediate findings of this project, the results contribute to a growing body of research on how online communities behave and how information spreads across social platforms, such as Reddit. Social media engagement is shaped by an extraordinarily complex interplay of psychological, temporal, and contextual factors. The strong influence of posting time, for example, aligns with circadian rhythm research showing that users exhibit predictable cycles of activity and emotional responsiveness throughout the day. The late-night spikes in engagement for emotionally charged topics reflect well documented patterns in digital behavior, where users tend to be more reactive, impulsive, and expressive during nighttime hours. This suggests that content strategy is not merely a matter of choosing the correct topic but also understanding the cognitive and emotional states of users at various times.

The dominance of topic as a predictor also reinforces the idea that content relevance and emotional salience are central to engagement. Posts that tap into controversy, misinformation, or strong opinions tend to generate more interaction, regardless of sentiment polarity. This is consistent with prior studies showing that emotionally provocative content spreads significantly more rapidly and widely than neutral or informational posts. While this can be leveraged for engagement optimization, it also raises ethical considerations about amplifying divisive or misleading narratives. The project therefore highlights the importance of balancing engagement goals with responsible content creation practices.

5.7 Technical Reflection and Model Behavior

From a technical standpoint, the Random Forest model performed as expected for a dataset with high variance in addition to nonlinear relationships. Tree-based models are particularly well suited for capturing interactions between categorical and temporal features, which explains their robust performance relative to linear models. The moderate R^2 score reflects the inherent unpredictability of social media engagement, where external factors such as trending events, user networks, and platform algorithms play significant roles that are not captured in the dataset. The MAE values indicate that while the model can approximate general engagement levels, extreme outliers remain difficult to predict. This is typical for engagement data, where viral posts behave as statistical anomalies.

The feature importance rankings also provide clear insight into how the model interprets the data. The relatively low importance of sentiment suggests that emotional tone alone is insufficient to predict engagement without considering topic context. This reinforces the idea that sentiment analysis should be paired with topic modeling or semantic analysis for more nuanced insights. The weak influence of month and year indicates that long term seasonality is minimal in this dataset, which is expected for discussions centered on vaccine myths which is a topic that remains relatively stable over time.

6. Recommendations and Conclusion

6.1 Actionable Recommendations

1. Post Topic 2 content late at night
2. Post Topic 3 content in the afternoon.
3. Avoid low engagement debate content.
4. Use Wednesday as the primary posting day.
5. Use LLMs to generate content variations.

6.2 Implementation Roadmap

1. Deploy engagement model
2. Integrate topic classifier
3. Automate posting schedule
4. Use LLMs for content generation.
5. Monitor performance and retain quarterly.

6.3 Expected Impact

- Higher overall engagement
- More efficient content production
- Better audience targeting
- Improved strategic planning

6.4 Future Work

- Incorporate image features.
- Use transformer-based sentiment models.
- Expand the dataset.
- Build a hybrid recommender system.

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